

DEUTSCHE BANK INDIA

Retail Banking — Digital Payments

BPMN PROCESS FLOWS

UPI Enablement Platform

Business Process Model and Notation Diagrams for Core UPI Workflows (TO-BE)

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1. Introduction

This document presents Business Process Model and Notation (BPMN)-style diagrams for the six core TO-BE workflows of the Deutsche Bank India UPI Enablement Platform. Each diagram uses colour-coded lanes to represent the actor or system responsible for each activity, standard BPMN shape conventions (circle = start event, double circle = end event, rounded rectangle = task, diamond = gateway/decision point), and solid/dashed arrows to distinguish primary flow from exception or conditional flow.

1.1 Notation Key

Symbol	Meaning
Circle (filled)	Start Event — the trigger that begins the process
Double Circle	End Event — a terminal outcome of the process
Rounded Rectangle	Task — a discrete unit of work performed by the actor/system in that lane
Diamond	Gateway — a decision point where the process branches based on a condition
Solid Arrow	Primary / success flow
Dashed Arrow	Alternate, exception, or conditional flow

Each diagram is traceable to its corresponding Use Case(s) and Functional Requirement IDs (FR-xx) defined in the Use Case Catalogue and Functional Requirements Specification (FRS).

2. UPI Registration Process

This process begins when a customer initiates UPI registration from the mobile app and covers mobile number verification, consent capture, account discovery, and UPI profile creation. The process terminates in either a successfully registered profile or a halted registration requiring remediation (e.g., mobile number mismatch or declined consent).

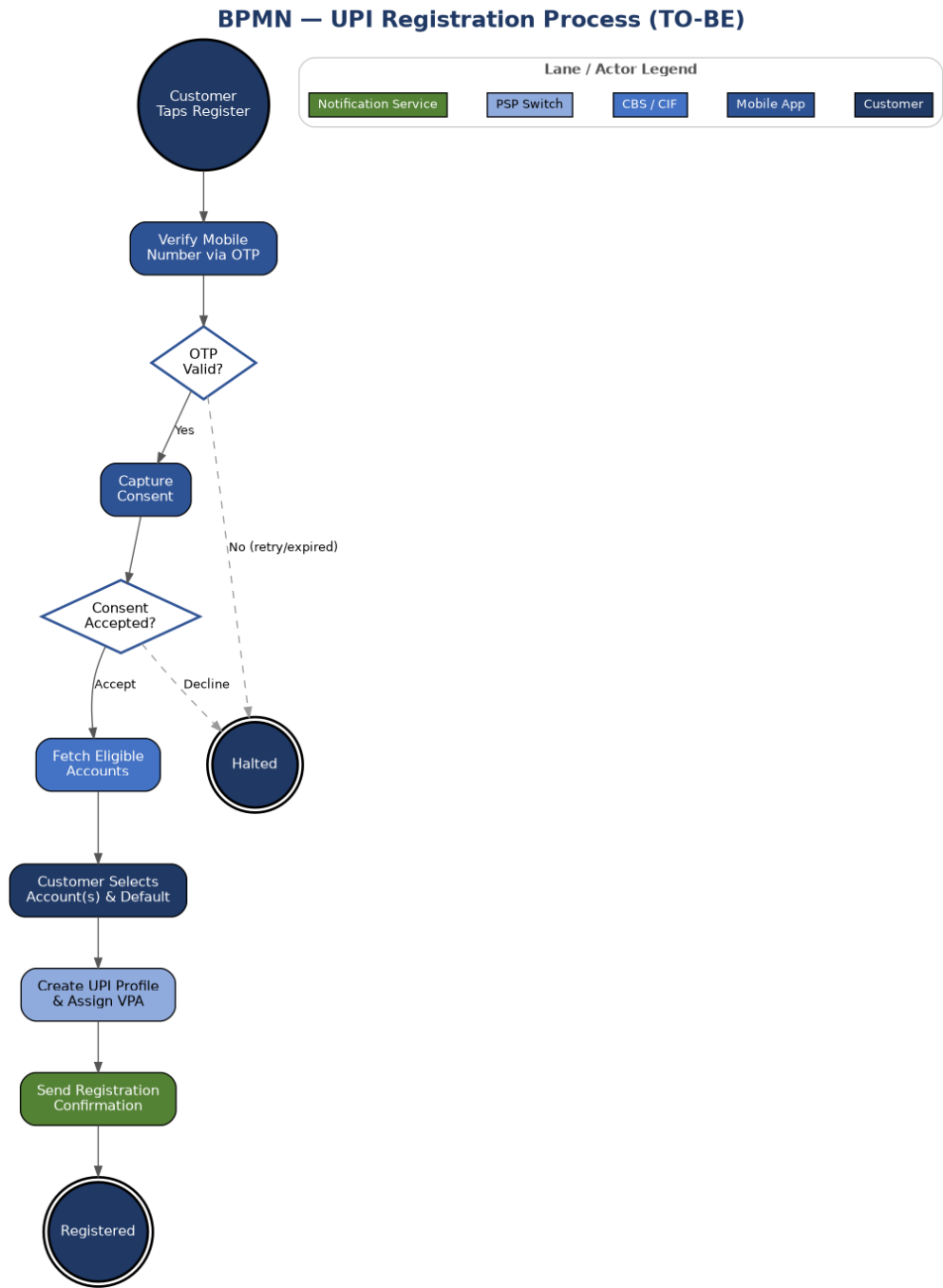


Figure 1 — BPMN: UPI Registration Process (TO-BE)

Related Use Cases / Requirements: UC-01, UC-02; FR-REG-01 to FR-REG-12

3. Bank Account Linking Process

This process covers the three core account-management actions available to a registered customer: linking a new eligible account, removing an existing linked account (with a guard against removing the default account without reassignment), and changing the default account.

BPMN — Bank Account Linking (Add / Remove / Set Default)

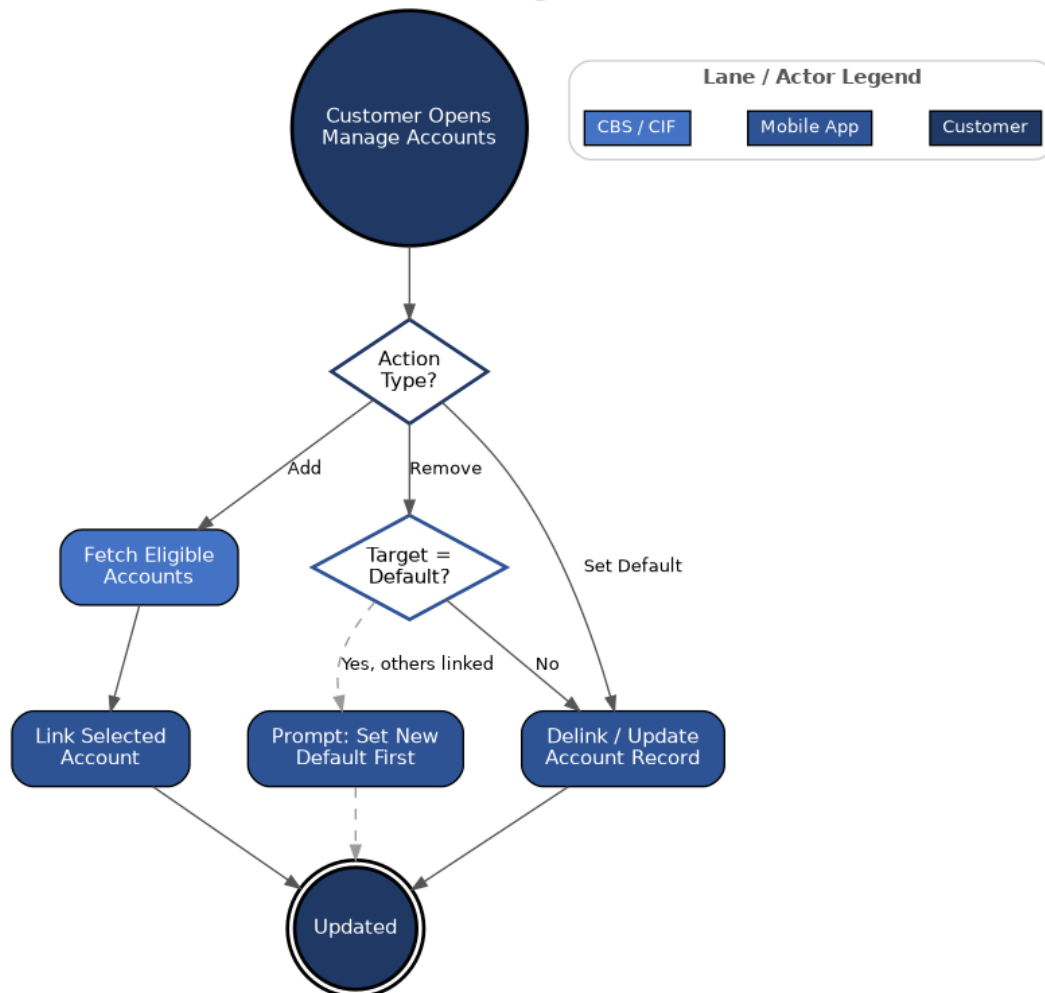


Figure 2 — BPMN: Bank Account Linking (Add / Remove / Set Default)

Related Use Cases / Requirements: UC-03, UC-04, UC-05; FR-ACC-01 to FR-ACC-08

4. UPI PIN Setup Process

This process covers PIN creation via either debit card validation or Aadhaar-OTP as an alternate NPCI-approved method, followed by mobile OTP verification and secure, on-device PIN encryption before submission to NPCI via the PSP switch.

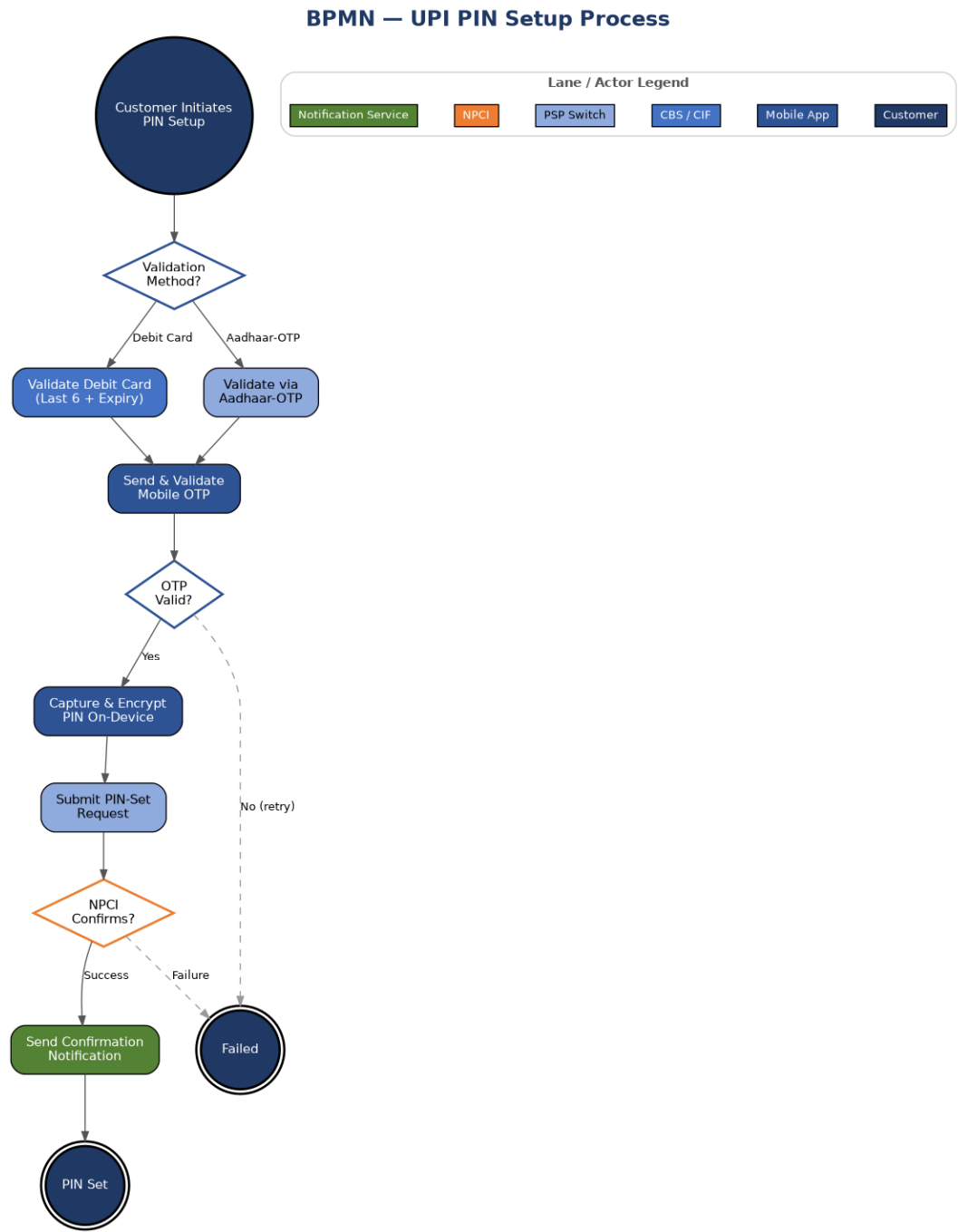


Figure 3 — BPMN: UPI PIN Setup Process

Related Use Cases / Requirements: UC-06; FR-PIN-01 to FR-PIN-04

5. Send Money Process

This is the platform's core transaction process, spanning payee resolution, balance validation, real-time fraud/velocity risk scoring, NPCI routing, and CBS debit/credit posting. The process has three distinct terminal outcomes: Success, Declined (blocked by risk or validation), and Reversed (where a timeout is later reconciled as a failed credit and automatically reversed).

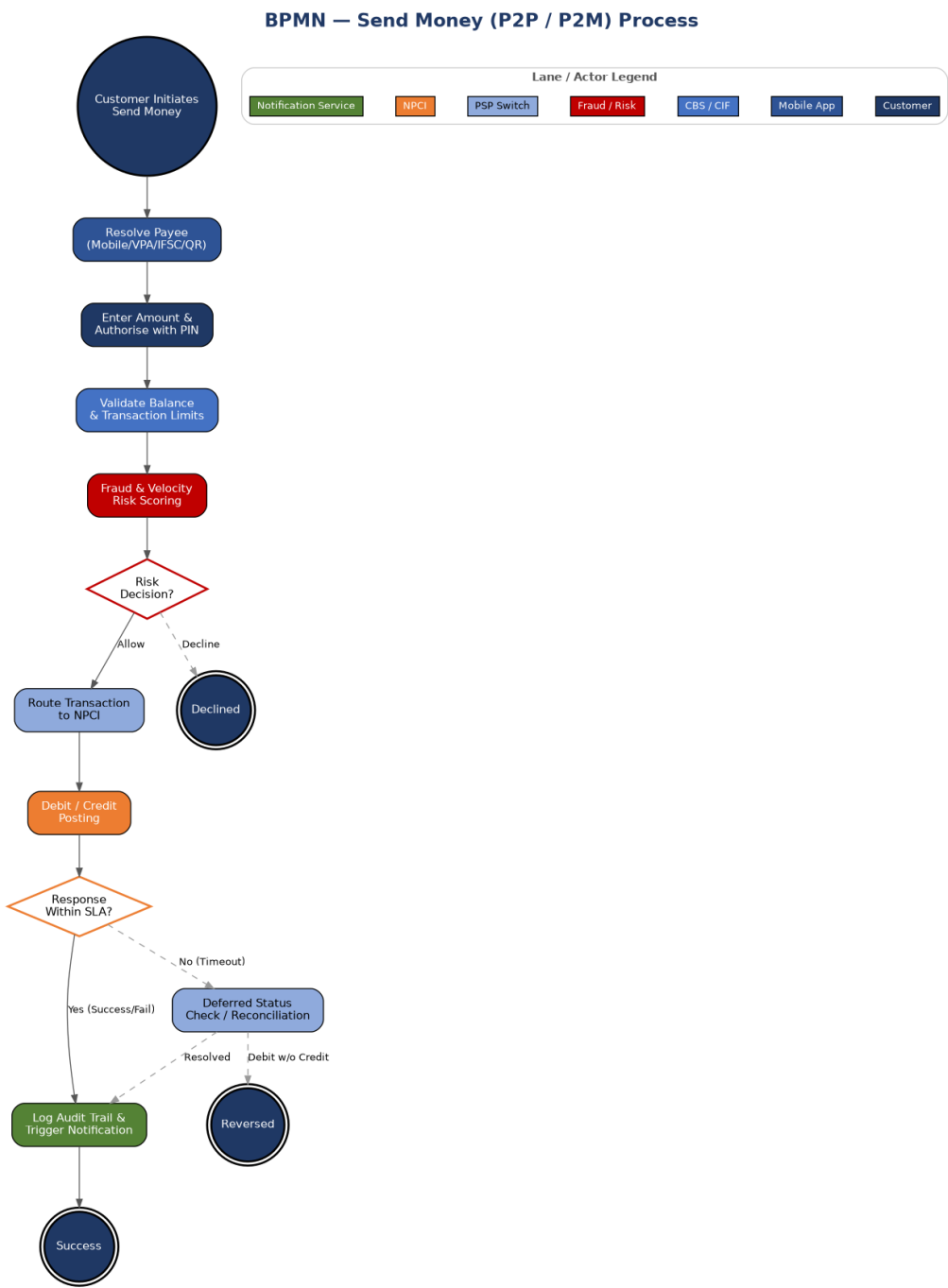


Figure 4 — BPMN: Send Money (P2P / P2M) Process

Related Use Cases / Requirements: UC-10 to UC-14; FR-TXN-01 to FR-TXN-14

6. Collect Request Process

This process covers the full collect-request lifecycle: creation by the requester, real-time notification to the payer, and the payer's decision to approve (triggering the standard Send Money sub-process), reject, or take no action, in which case the request is automatically expired by a scheduled system job after 24 hours.

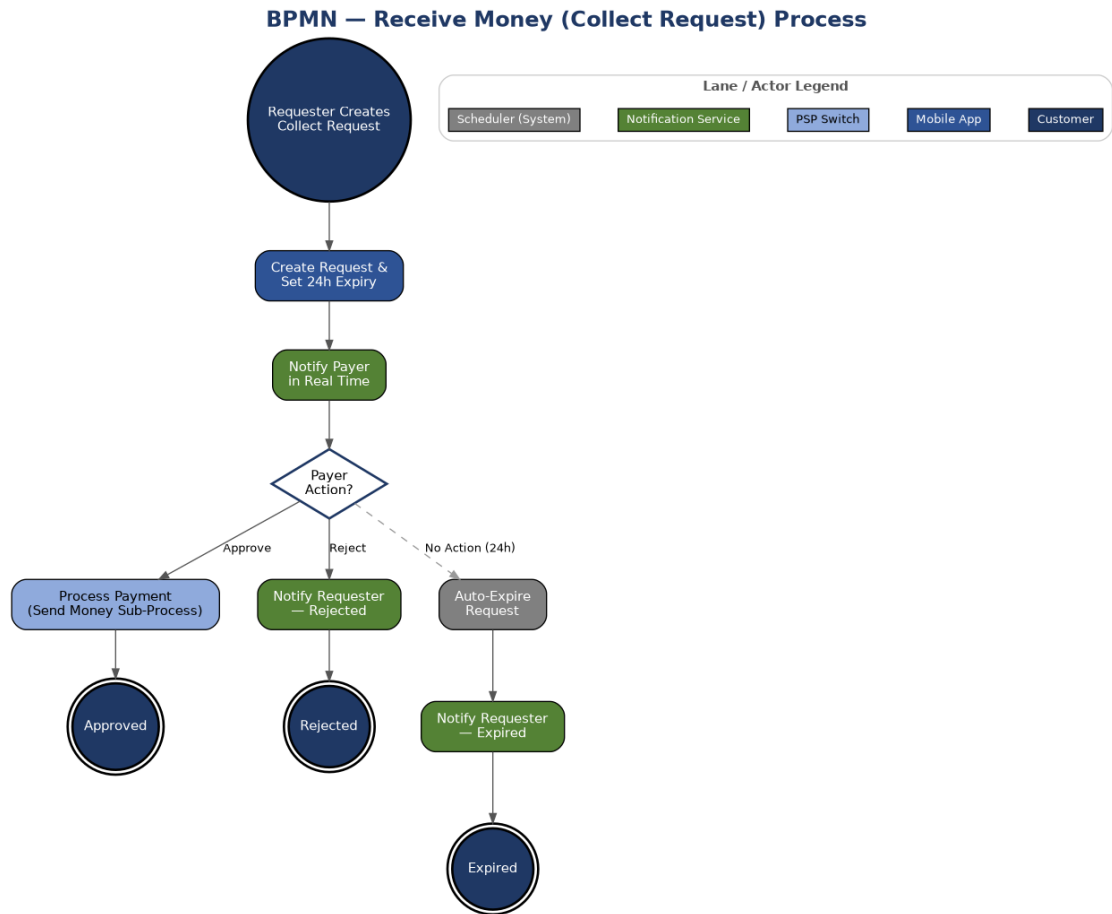


Figure 5 — BPMN: Receive Money (Collect Request) Process

Related Use Cases / Requirements: UC-15 to UC-18; FR-COL-01 to FR-COL-08

7. Dispute Management Process

This process covers dispute intake, conditional escalation to NPCI depending on dispute category, Operations investigation, automatic TAT-breach escalation via a scheduled job, and final resolution as either a processed refund or a rejected dispute with reason, each ending in a customer notification.

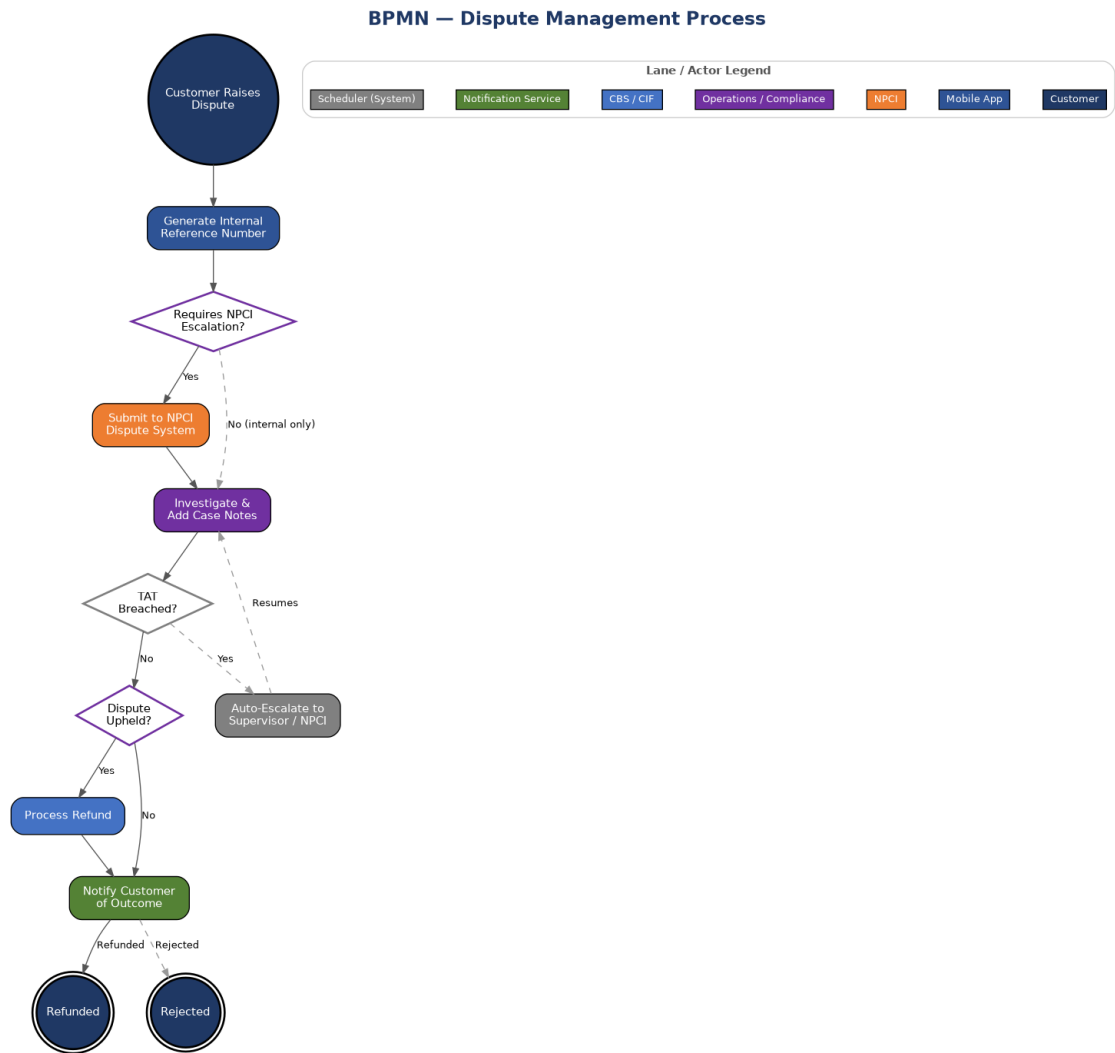


Figure 6 — BPMN: Dispute Management Process

Related Use Cases / Requirements: UC-26 to UC-29; FR-DIS-01 to FR-DIS-08